IMY 761 Deliverable 7

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Overview

For this deliverable, you will write up the details and results of your data collection and analysis. This will include providing the details of how the data were collected, providing the data itself and discussing the significance thereof as it pertains to the research questions, and providing the details of the analysis done to answer each research question.

Practical instructions
Do not create a new document. Fill everything in on this document and submit it in a .docx format.

Exercise 1: Revisit your research questions

Include your latest research questions here. Make sure to mark the primary and sub-questions. Also, ensure that you indicate how each question will be answered. Those questions intended to be answered through the empirical work will be used in the remainder of this document.

Main question: The Effect of Diegetic Versus Non-Diegetic Failure Feedback on Player Frustration and Perceived Competence in a 2D Platformer Game.

Sub-questions

Question	**How it will be answered**
How do existing definitions distinguish diegetic from non-diegetic game interfaces and feedback?	Through literature: a review of how videogame research defines diegetic and non-diegetic interfaces, and the frameworks used to tell them apart.
How does interface diegesis compare to non-diegesis in terms of its effect on immersion, frustration, and perceived competence, according to existing research?	Through literature: a review of previous empirical/quantitative studies comparing diegetic and non-diegetic interfaces and feedback.
How does diegetic failure feedback compare to non-diegetic failure feedback in terms of players' self-reported frustration and perceived competence? This one is answered by the questionnaire and t-tests.	Empirically. Post-play questionnaire using BANGS competence satisfaction and frustration items and GEQ tension/annoyance items.

	The scores are compared between conditions with exploratory independent-samples t-tests.
How do players' descriptions of failure, competence, and motivation to continue compare between the diegetic and non-diegetic conditions? This one is answered by the interviews.	Empirically. Semi-structured post-play interviews, analysed with Braun and Clarke's reflexive thematic analysis and supported by observation notes of comments players made while playing.

Exercise 2: Details of the data collection

Provide full details of how all data were collected. This includes the instruments used, how they were administered, the format of collected data, and how they were processed for analysis.

This should be 150-250 words:

The Data Collection

Data was collected from ten participants, with five assigned to each game condition. In both versions, an animal companion followed the player. In the diegetic condition, it showed the player how to get past an obstacle after they failed, while in the non-diegetic condition it did not react and an on-screen arrow and text gave the feedback. Participants were allocated as they became available. Seven sessions were conducted in person using the researcher's MacBook, while three were conducted remotely through video calls. Remote participants received the game through Google Drive, using either a macOS or Windows build, and shared their screens during gameplay.

All sessions followed the same researcher script. Participants were provided with an information sheet and gave consent before playing the five-level 2D platformer independently for up to 15 minutes.

Following gameplay, participants completed a nine-item questionnaire through Google Forms. Items 1-6 were drawn from the competence subscales of the Basic Needs in Games Scale (BANGS), rated from 1 to 7, while items 7-9 measured tension/annoyance using the Game Experience Questionnaire (GEQ), rated from 0 to 4. Responses were anonymised using participant numbers, with condition assignments recorded separately.

Participants then completed a semi-structured interview consisting of five open-ended questions. Interviews were audio-recorded with consent, automatically transcribed and anonymised. During gameplay, the researcher also typed brief notes of participants' comments.

Questionnaire responses were exported to a spreadsheet, where mean scores for competence satisfaction (items 1, 3, 5), competence frustration (items 2, 4, 6) and tension/annoyance (items 7-9) were calculated. Differences between conditions were examined using independent-samples t-tests and effect sizes, while interview transcripts were retained for qualitative analysis of participants' experiences of failure and feedback.

Step 1: The data

The result of doing data collection is, of course, the data itself that need to be analysed. Before jumping into the analysis, consider the data and its relevance within your study. Understanding how the data relate to your study and allow you to answer your research questions will, in turn, inform how they should be analysed.

The following example provides two sub-questions from the same study intended to be answered through empirical data. It includes the description of the data used in the study and discusses the significance of said data in the study context.

Research sub-question 1:

How does the use of VR affect the perceived cognitive load of visualised data?

Data (not included here): results from a cognitive load survey administered to both the control group (using the PC version of the application) and the test group (using the VR version).

The data provided allow us to answer this sub-question by comparing the scores given for perceived cognitive load for the control and test groups. Using this comparison, we can determine whether VR led to decreased/increased cognitive load or had no significant effect on cognitive load during the test.

Research sub-question 2:

What are the salient features of VR that factor into user preferences in the context of data visualisation?

Data (not included here): transcriptions of short interviews conducted with participants from both groups.

The interview data allow us to identify features that participants experienced during the test which affected their overall experience of either version of the application. By analysing the data for recurring themes, we can identify which features affected participants' experience and whether this experience was positive/negative/neutral in the context of the task provided.

Exercise 3

Follow the sample above for your own study. For each research to be answered empirically (including those to be answered in combination with the literature), provide the question, the relevant raw data, and a short discussion of the significance of the data toward answering the question.

The raw data can be provided as linked Google Sheets or Google Docs files (remember to check sharing settings and to anonymise the data). Include only the data which is relevant to answer each question. Do not simply include all the data that has been collected if it does not pertain to the question:

Research sub-question 3:

How does diegetic failure feedback compare to non-diegetic failure feedback in terms of players' self-reported frustration and perceived competence?

Data (Post-Play Questionnaire (Responses)**):** post-play questionnaire responses from both the diegetic and non-diegetic groups. Each participant rated six BANGS items on a 1-7 scale (competence satisfaction and competence frustration) and three GEQ tension/annoyance items on a 0-4 scale.

The scores of the two groups can be compared across three subscales: competence satisfaction, competence frustration, and tension/annoyance. Each participant's score can be calculated by taking the average of the three items in each subscale. The group averages can then be compared using exploratory independent-samples t-tests. This is so that we can see whether diegetic feedback is associated with higher, lower or similar levels of frustration and perceived competence compared to non-diegetic feedback. Competence satisfaction and competence frustration can be analysed separately because BANGS treats them as different measures. This makes it possible to see whether the type of feedback is associated with how capable players felt, how incapable they felt, or both.

Research sub-question 4:

How do players' descriptions of failure, competence, and motivation to continue compare between the diegetic and non-diegetic conditions?

Data (submit_transcripts**):** anonymised transcripts of the semi-structured post-play interviews with participants from both groups, together with typed observation notes of comments participants made while playing.

The interview responses can be analysed to understand how participants experienced failure in their version of the game. This includes whether they understood what went wrong, how capable they felt of improving, and what encouraged them to continue playing or stop. The interview transcripts can be analysed using Braun and Clarke's reflexive thematic analysis to identify recurring themes. These themes are then compared between the two groups to see whether the type of feedback influences how players described their frustration and competence. The observation notes can also be used to capture how participants reacted

while playing, including comments about how useful the companion was or how fair they thought certain hazards were.

Step 2: Data analysis

The next step is to analyse the data to answer each research sub-question. The choice of analysis method depends on the type of data and the goal of the analysis.

Answering sub-question 1:

The survey data provided above would be analysed using a t-test to determine whether there is a significant difference between the measured outcome for the two groups (i.e., the control group and the test group). This would allow us to say whether there is a difference and in which direction, i.e., whether cognitive load is higher in which intervention (PC or VR).

How does the use of VR affect the perceived cognitive load of visualised data?
It was hypothesised that the VR group would experience a greater cognitive load than the control group. Therefore, a one-tailed independent samples t-test was performed to compare cognitive load in the control group and the VR group.

There was a significant difference in cognitive load between the control group (M = [Mean], SD = [standard deviation]) and the VR group (M = [Mean], SD = [standard deviation]); t(df) = [t-value], p = [p-value]. The results show that the VR group experienced a significantly greater cognitive load than the control group when interacting with the visualised data.

Answering sub-question 2:

The interview data provided above will be analysed using thematic analysis to provide a list of themes in the data. The links between relevant themes can also be explored (e.g., screen resolution might be linked with text legibility). Using diagrams would be a useful way to point out themes and sub-themes and to indicate links between themes.

HINT: some validated instruments include a guide for analysing the data, so it makes sense to adhere to this where available.

Exercise 4

For each research to be answered empirically (including those to be answered in combination with the literature), provide the question, describe the analysis performed, and the analysis results. Use the data analysis to provide a clear, concrete answer for each research question.

Note: for this deliverable, you are not required to answer the sub-questions, which will only be answered through the literature. You are also not required to answer the primary

research question. In the discussion section of your final article, each of the sub-questions will be answered and then these answers will be combined to answer the primary question.

Answering sub-question 3:

How does diegetic failure feedback compare to non-diegetic failure feedback in terms of players' self-reported frustration and perceived competence?

The post-play questionnaire responses were scored following the guidance for each instrument. For each participant, three subscale scores were calculated as the mean of their items:

I. Competence satisfaction (BANGS items 1, 3 and 5; scale 1-7)
II. Competence frustration (BANGS items 2, 4 and 6; scale 1-7)
III. Tension/annoyance (GEQ items 7, 8 and 9; scale 0-4)

Competence satisfaction and competence frustration were kept as separate constructs. Each subscale was compared between the diegetic group (n = 5) and the non-diegetic group (n = 5) using a two-tailed independent-samples t-test, with a significance level of α = .05. Cohen's d is reported as the effect size. Due to the sample size being small, these tests were exploratory.

Results

Subscale	Diegetic Mean (Standard Deviation)	Non-diegetic Mean (Standard Deviation)	t(df)	p	d
Competence satisfaction (1-7)	6.07 (1.01)	6.00 (1.35)	t(7.40) = 0.09	.932	0.06
Competence frustration (1-7)	2.40 (0.98)	2.53 (1.22)	t(7.66) = -0.19	.854	-0.12
Tension/annoyance (0-4)	1.33 (0.85)	1.73 (1.30)	t(6.89) = -0.58	.583	-0.36

Table 1. Subscale scores by condition and results of the two-tailed independent-samples t-tests (n = 5 per group).

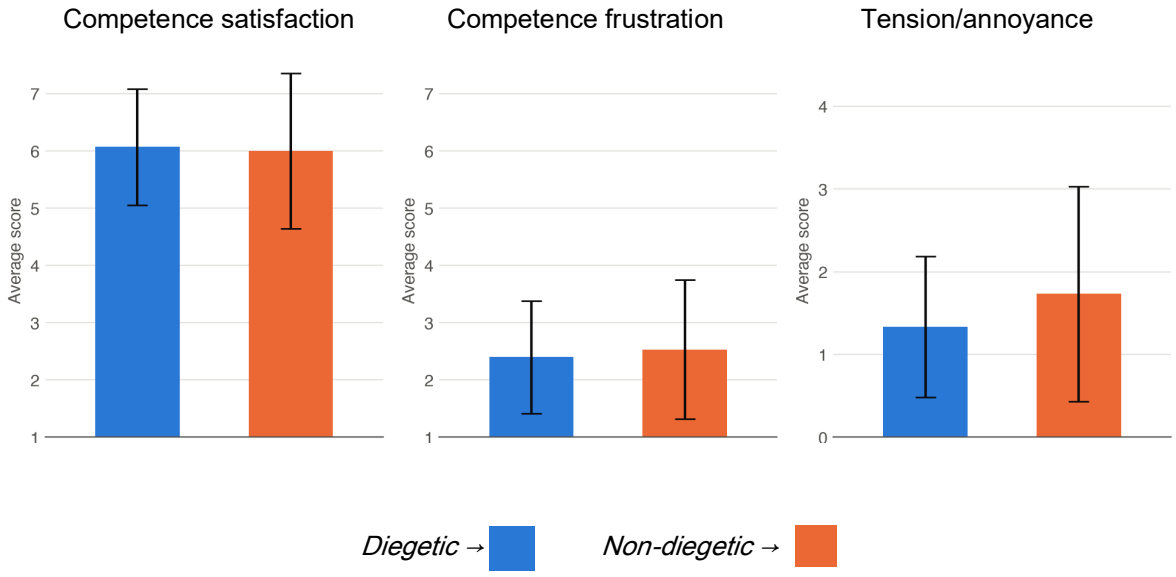


Figure 1. Average scores for the diegetic (n = 5) and non-diegetic (n = 5) groups. Error bars show ±1 standard deviation. There were no significant differences between the groups on any measure (all $p > .05$).

Figure 1 shows the average score of each group on the three subscales:

- There was no significant difference in competence satisfaction between the diegetic group ($M = 6.07$, $SD = 1.01$) and the non-diegetic group ($M = 6.00$, $SD = 1.35$); $t(7.40) = 0.09$, $p = .932$, $d = 0.06$.

- There was no significant difference in competence frustration between the diegetic group ($M = 2.40$, $SD = 0.98$) and the non-diegetic group ($M = 2.53$, $SD = 1.22$); $t(7.66) = -0.19$, $p = .854$, $d = -0.12$.

- There was no significant difference in tension/annoyance between the diegetic group ($M = 1.33$, $SD = 0.85$) and the non-diegetic group ($M = 1.73$, $SD = 1.30$); $t(6.89) = -0.58$, $p = .583$, $d = -0.36$.

In this sample, diegetic and non-diegetic failure feedback produced similar levels of self-reported perceived competence and frustration. Both groups reported high competence satisfaction and low competence frustration. Players who received diegetic feedback reported slightly lower tension/annoyance than those who received non-diegetic feedback, a small effect ($d = -0.36$), but the difference was not statistically significant. With five participants per group, the study could only reliably detect very large differences, so these results show no clear effect of feedback type.

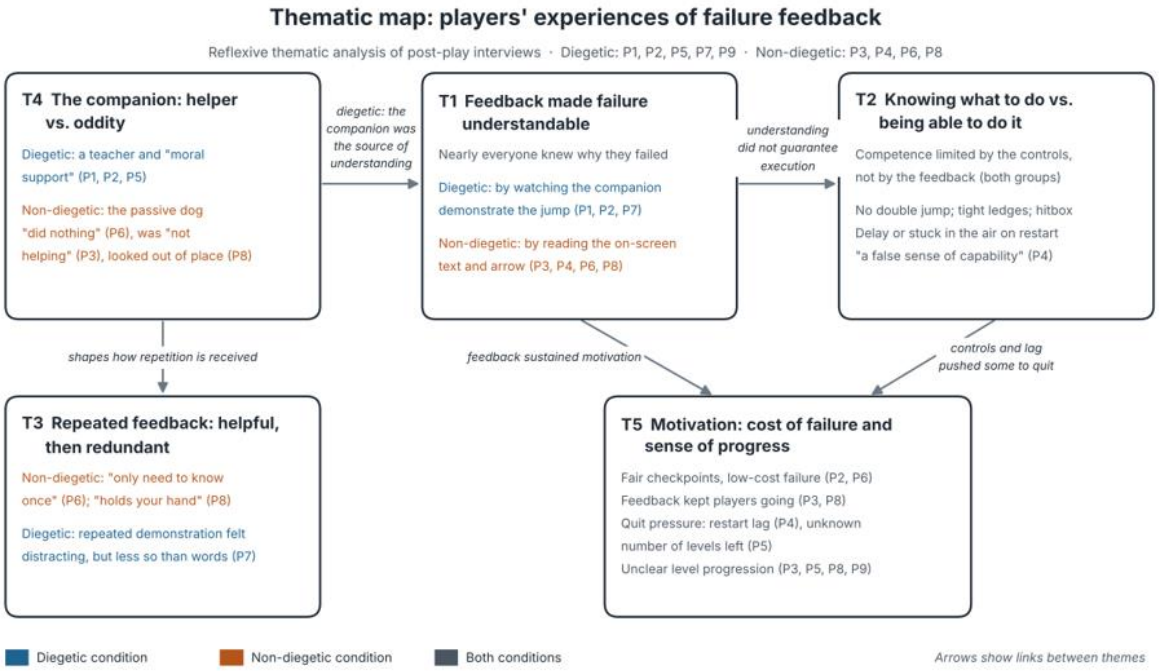
Answering sub-question 4:

How do players' descriptions of failure, competence, and motivation to continue compare between the diegetic and non-diegetic conditions?

The interview transcripts were analysed using Braun and Clarke's reflexive thematic analysis. The researchers first familiarised themselves with the data by reading each transcript alongside the audio and correcting transcription errors. Segments relating to failure, understanding of feedback, competence and motivation were then coded inductively. The codes were grouped into candidate themes, which were reviewed against the full dataset, then defined and named. Finally, the themes were compared between the diegetic group (P1, P2, P5, P7, P9) and the non-diegetic group (P3, P4, P6, P8, P10), where 'P' refers to 'Participant', to see where the two groups' accounts were similar and where they differed.

Results

Five themes were developed. These are summarised in Figure 2, which also shows how the themes link to each other.



Theme 1: Feedback made failure understandable in both conditions.

Almost every participant said that they knew why they had failed. In the diegetic condition, this understanding came from watching the companion. P1 initially wondered why they kept dying, but understood once they saw the animal jump over the hazard. P2 said the dog showing what to do made it “easy to understand and easy to follow.” In the non-diegetic condition, understanding came from reading the on-screen text and arrow. P3 said the description at the top explained what went wrong, and P8 said the game tells you exactly what you did wrong. The type of feedback therefore changed how players understood failure, but not whether they understood it.

Theme 2: Knowing what to do was not the same as being able to do it.

Participants in both groups described their competence as limited by the controls rather than by the feedback. Players expected a double jump that did not exist, found the landing areas and hitbox hard to judge, and noticed delays or the character getting stuck in the air when restarting. P3 said they knew what to improve but “it was the implementation that was difficult.” P6 described failure as “a physical input point” (or in other words, the controls were the issue) rather than a lack of understanding. P4 felt the feedback gave them “a false sense of capability,” because it told them to jump further when the controls did not allow it. Essentially, frustration was mostly directed at the spikes and controls, not at the feedback itself, which participants generally described as “valid” or “helpful.”

Theme 3: Repeated feedback was helpful at first, then redundant.

Several participants felt they only needed the feedback once. This came through most strongly in the non-diegetic group. P6 rated the text as clear and helpful but then distracting, because “I only need to know once.” P8 said the game “holds your hand,” so it stopped

being a mental exercise. P3 disagreed, saying it was good that the feedback was constantly there as a reminder. In the diegetic group, P7 described the companion's repeated demonstration as clear and then distracting.

Theme 4: The companion was a helper when it demonstrated, and an oddity when it did not.

This was the clearest difference between the groups. Diegetic participants described the companion as a teacher or supporter. P1 liked that the companion would show you how to do it, P2 credited the companion with making them want to keep playing, and P5 called it "moral support". Non-diegetic participants, whose companion did not demonstrate anything, described it as having no purpose. P6 said it "did nothing." P3 felt it was "unfair" that the dog was "not helping" and joked that it should have died with them. P8 found it cute but visually out of place.

Theme 5: Motivation to continue depended on the cost of failure and a sense of progress.

Most participants wanted to keep playing after failing. They described the checkpoints as fair and failure as low-cost: P2 said failing "didn't feel catastrophic," and P6 said there were "no consequences." Some linked their motivation directly to the feedback. P3 said that without feedback they might have felt demotivated, and P8 said that they "probably would have quit" if the arrow had not shown them that there was no double jump. Demotivation came from other sources. For P4, it was the lag when restarting, saying that "it was easier to quit". For P5, it was not knowing how many levels remained ("Am I just gonna fail forever?"), which was a point brought up by most participants.

Diegetic participants described failure more positively (such as "motivated to try again,") and often credited the companion for it. Non-diegetic participants gave more mixed accounts. They were more likely to call repeated feedback redundant, and saw the passive companion as pointless.

The individual accounts also matched the questionnaire scores. P4, who described feeling "very demotivated," had the highest competence, frustration and tension/annoyance scores in the sample. P8, who felt the feedback kept them on track, had the lowest competence frustration.

Both kinds of feedback (questionnaires and semi-structured interviews) did the same job of showing players what went wrong, so competence scores were similar. However, the diegetic presentation was experienced as slightly less irritating, which matches the insignificant and small difference in annoyance.

Results & discussion

Criteria	Poor	Average	Good	Excellent
Significance: The significance of each dataset is clearly motivated in clearly aligned to answer the sub RQ	1,2	3	4	5
Analysis: Choice of analysis method is aligned to the data and intended outcomes.	1	2	3	
Analysis is performed correctly.	1, 2	3	4	5
Analysis is used to provide answers to the RQ.	1, 2	3	4	5
Total	/18			